The Physics Notebook

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1 Prerequisite Mathematics

§1.1 Vectors

Vectors are quantities with magnitude and direction, contrastingly to scalars which only have magnitude. For our purposes, vectors can be understood as directed line segments, although they're more rigorously defined as elements of a vector space, or can be understood through rotations. We'll talk about rotations later, but will not deal with vector spaces right now.

I shall use boldface to denote vectors, such as $\boldsymbol{A}$.

The utility of vectors stem from their independence from a particular coordinate system. They're directed *line segments*, only their length matters and we're not concerned with their end points. This means that a vector can be freely translated.

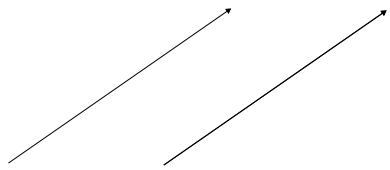

Figure 1.1: Identical Vectors

The magnitude of a vector, its norm, is its euclidean "length", we denote it by $\|\boldsymbol{A}\|$ or simply A.

A unit vector, $\hat{\boldsymbol{a}}$ ("a hat") is a vector whose magnitude is unity. We use the unit vector to often denote the direction of a vector by multiplying the unit vector with its magnitude. For instance, the unit vector that point is the direction of A, $\hat{\boldsymbol{A}}$ can be calculated as,

$$\hat{\boldsymbol{A}} = \frac{\boldsymbol{A}}{A}$$

where we divide by the magnitude of A to ensure that the magnitude of $\hat{\boldsymbol{a}}$ is 1.

§1.2 Vector Algebra

There are two operations generally defined on vectors, scalar multiplication and vector addition.

§1.2.1 Scalar Multiplication

Multiplying a vector $\boldsymbol{A}$ by a scalar c results in another vector, $c\boldsymbol{A}$ parallel or antiparallel(pointing in the opposite direction) to $\boldsymbol{A}$ scaled by a factor of $|c|$.

If c is positive, then the resultant vector is simply scaled by it. If we multiply a vector by -1, then the result is the simple reversal in direction of vectors. If c is negative, then the vector is reversed and scaled by $|c|$.

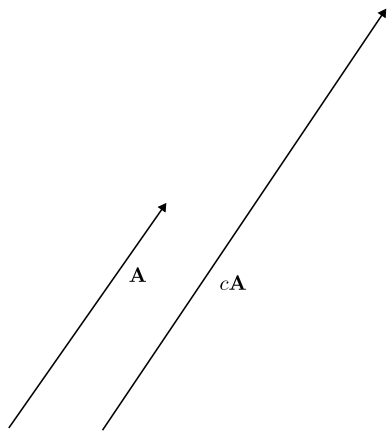

Figure 1.2: Scalar Multiplication

§1.2.2 Vector Addition

We define an operation called addition of vectors which produces another vector. To add two vectors, $\boldsymbol{A}$ and $\boldsymbol{B}$, we stick the tail of one vector with the head of another.

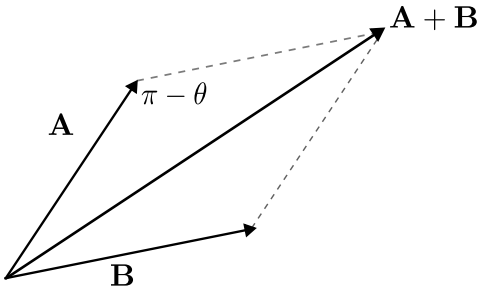

Figure 1.3: Vector Addition

We can calculate the magnitude of their resultant vector, $\|\boldsymbol{A}+\boldsymbol{B}\|$ by using the cosine rule. In fig. 1.3 θ is the angle between $\boldsymbol{A}$ and $\boldsymbol{B}$.

$$\begin{aligned}\|\boldsymbol{A}+\boldsymbol{B}\|^2 &= A^2+B^2-2AB\cos(\pi-\theta)\\ \|\boldsymbol{A}+\boldsymbol{B}\|^2 &= A^2+B^2+2AB\cos\theta\\ \|\boldsymbol{A}+\boldsymbol{B}\| &= \sqrt{A^2+B^2+2AB\cos\theta}\end{aligned}$$

To subtract two vectors, we simply reverse the vector and then add the resulting vector.

Example 1.1 Consider the infinitesimal change vector, $d\boldsymbol{A} = \boldsymbol{A}(t+dt) - \boldsymbol{A}(t)$.

We have,

$$\begin{aligned}d\boldsymbol{A} &= A(t+dt)\hat{\boldsymbol{A}}(t+dt) - A(t)\hat{\boldsymbol{A}}(t)\\ d\boldsymbol{A} &= A(t+dt)\hat{\boldsymbol{A}}(t+dt) - A(t+dt)\hat{\boldsymbol{A}}(t)\\ &\quad + A(t+dt)\hat{\boldsymbol{A}}(t) - A(t)\hat{\boldsymbol{A}}(t)\\ d\boldsymbol{A} &= A(t+dt)(\hat{\boldsymbol{A}}(t+dt)-\hat{\boldsymbol{A}}(t)) + \hat{\boldsymbol{A}}(t)(A(t+dt)-A(t))\\ d\boldsymbol{A} &= A\,d\hat{\boldsymbol{A}} + dA\hat{\boldsymbol{A}}\end{aligned}$$

The first term represents a change in direction and the second represents a change in magnitude.

The magnitude of the first can be calculated as,

First consider fig. 1.4.

Now, let us first work out a geometric argument for $A\,d\hat{\boldsymbol{A}}$ which is perpendicular to $\boldsymbol{A}$. In the limiting case, $A\,d\hat{\boldsymbol{A}}$, the change in magnitude is negligible, we only observe a change in direction. For now, let us consider $\Delta\boldsymbol{A}$ perpendicular to $\boldsymbol{A}$ as an approximation.

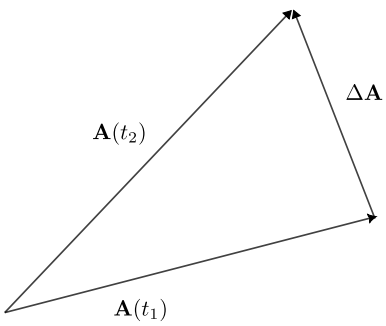

Figure 1.4: $\Delta\boldsymbol{A} \perp \boldsymbol{A}$

This allows us to consider a simple geometric argument which gives $\|\Delta\boldsymbol{A}\| = 2A\sin(\theta/2)$ where θ is the angle between $\boldsymbol{A}(t_1)$ and $\boldsymbol{A}(t_2)$.

Since we're considering values of small θ, we may use the small angle approximation, $\sin(\theta) \approx \theta$. Therefore,

$$\|\Delta\boldsymbol{A}\| \approx 2A \times \frac{\theta}{2} \tag{1.1}$$

$$\approx A \times \theta \tag{1.2}$$

Which is equivalent to $A\Delta\theta$. In the limit $\Delta\theta \to 0$,

$$dA_\perp = A\,d\theta \tag{1.3}$$

Which finally gives us,

$$\left\|\frac{d\boldsymbol{A}_\perp}{dt}\right\| = A\frac{d\theta}{dt} \tag{1.4}$$

which is the magnitude of $A\,d\hat{\boldsymbol{A}}$. Thus,

$$\|d\hat{\boldsymbol{A}}\| = \frac{d\theta}{dt} \tag{1.5}$$

§1.3 Dot Product

The dot product of two vectors results in a scalar and is defined as,

$$\boldsymbol{A} \cdot \boldsymbol{B} = AB \cos\theta$$

where θ is the angle between $\boldsymbol{A}$ and $\boldsymbol{B}$.

Note that the quantity $\boldsymbol{A} \cdot \boldsymbol{B}$ is just A times the magnitude of the projection of $\boldsymbol{B}$ on $\boldsymbol{A}$. Similarly, $\boldsymbol{B} \cdot \boldsymbol{A}$ is magnitude of the projection of $\boldsymbol{A}$ on $\boldsymbol{B}$ times B.

We may note that,

$$\boldsymbol{A} \cdot \boldsymbol{A} = A^2$$

Or,

$$A = \sqrt{\boldsymbol{A} \cdot \boldsymbol{A}}$$

It can be easily shown that the scalar product distributes over vector addition. Another thing of importance is the identity(which is also easy to derive),

$$\frac{d\boldsymbol{A} \cdot \boldsymbol{B}}{dt} = \frac{d\boldsymbol{A}}{dt} \cdot \boldsymbol{B} + \frac{d\boldsymbol{B}}{dt} \cdot \boldsymbol{A}$$

§1.4 Cross Product

The cross product is another product operation on vectors, this product produces a vector.

It is defined as,

$$\boldsymbol{A} \times \boldsymbol{B} = AB \sin\theta \hat{\boldsymbol{n}}$$

Where $\hat{\boldsymbol{n}}$ is the vector perpendicular to the plane containing $\boldsymbol{A}$ and $\boldsymbol{B}$. Since there are two directions that $\hat{\boldsymbol{n}}$ can have, we define the two vectors and their cross product to form a right-hand triple.

Place your fingers in the direction of $\boldsymbol{A}$ and curl them, along the smaller angle, towards $\boldsymbol{B}$. Then the direction in which your thumb is pointing is the direction of $\boldsymbol{A} \times \boldsymbol{B}$.

§1.5 Vectors in Component form

Although we have worked without co-ordinate systems till now, to actually extract any meaning from vector operations, a coordinate system is necessary though the concept it describes is actually independent of any coordinate system. A spatial, three-dimensional coordinate system essentially establishes three basis vectors.

Basis vectors are vectors that are linearly independent. That is, their linear combination,

$$c_1\boldsymbol{e}_1 + c_2\boldsymbol{e}_2 + \cdots + c_n\boldsymbol{e}_n$$

only has a trivial solution of $c_1 = c_2 = \cdots = c_n = 0$, when equated to the n dimensional null vector, $c_1\boldsymbol{e}_1 + c_2\boldsymbol{e}_2 + \cdots + c_n\boldsymbol{e}_n = \boldsymbol{0}$.

These vectors being linearly independent imply that they can not be written as linear combination of other vectors. Otherwise, we could have simply written the term $\boldsymbol{e}_i$ as the linear combination of other vectors whenever we needed to use it.

Although it might not be so obvious, all orthogonal sets of vectors are linearly independent. If we represent these as $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n\}$, we can construct a set of basis vectors that are not orthogonal as, $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_{n-1}, \boldsymbol{e}_1 + \boldsymbol{e_2} + \cdots + \boldsymbol{e}_n\}$.

Which means that we don't always need the basis vectors to be orthogonal, but in practice we usually adopt orthogonal basis vectors.

In particular, we'll use the basis vectors which have magnitude of 1, i.e. unit vectors.

In cartesian co-ordinates, we use the unit vectors along the three axes, x, y and z. These are written as $\hat{\boldsymbol{i}}$, $\hat{\boldsymbol{j}}$ and $\hat{\boldsymbol{k}}$ respectively.

But how can we write an arbitrary vector $\boldsymbol{A}$ in terms of these unit vectors? The answer lies in projecting the vector along the axes. We use the dot product for it.

$$A_x = \boldsymbol{A} \cdot \hat{\boldsymbol{i}} = A \cos \alpha \tag{1.6}$$
$$A_x = \boldsymbol{A} \cdot \hat{\boldsymbol{j}} = A \cos \beta \tag{1.7}$$
$$A_x = \boldsymbol{A} \cdot \hat{\boldsymbol{k}} = A \cos \gamma \tag{1.8}$$

Where α, β, and γ are the angles the vector makes with the x, y and z axes respectively. The cosine of these angles are called directional cosines.

Thus, we write $\boldsymbol{A}$ in terms of it elements, in matrix form, as,

$$\boldsymbol{A} = \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \tag{1.9}$$

Now scalar multiplication is simply,

$$c\boldsymbol{A} = \begin{pmatrix} cA_x \\ cA_y \\ cA_z \end{pmatrix}$$

And vector addition is,

$$\boldsymbol{A} + \boldsymbol{B} = \begin{pmatrix} A_x + B_x \\ A_y + B_y \\ A_z + B_z \end{pmatrix}$$

For the dot product, representing $\boldsymbol{A}$ and $\boldsymbol{B}$ in terms of the basis vectors,

$$\boldsymbol{A} \cdot \boldsymbol{B} = (A_x\hat{\boldsymbol{i}} + A_y\hat{\boldsymbol{j}} + A_z\hat{\boldsymbol{k}}) \cdot (B_x\hat{\boldsymbol{i}} + B_y\hat{\boldsymbol{j}} + B_z\hat{\boldsymbol{k}})$$
$$\boldsymbol{A} \cdot \boldsymbol{B} = A_xB_x + A_yB_y + A_zB_z$$

Where we use the distributive property of dot product and the fact that $\hat{\boldsymbol{i}} \cdot \hat{\boldsymbol{j}} = \hat{\boldsymbol{j}} \cdot \hat{\boldsymbol{k}} = \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{i}} = 0$.

Finally, the cross product is a little difficult to compute. It can be computed either as the determinant,

$$\boldsymbol{A} \times \boldsymbol{B} = \begin{vmatrix} \hat{\boldsymbol{i}} & \hat{\boldsymbol{j}} & \hat{\boldsymbol{k}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Or using the matrix notation,

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \times \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} = \begin{pmatrix} A_y B_z - A_z B_y \\ A_z B_x - A_x B_z \\ A_x B_y - A_y B_x \end{pmatrix}$$

To compute a particular component, cover up that particular row. Then, multiply the component of $\boldsymbol{A}$ in the row below (loop to the top if necessary) with the diagonal component of $\boldsymbol{B}$ and subtract it by the product of the component of $\boldsymbol{A}$ two rows below, and the diagonal component of $\boldsymbol{B}$. For example, for the y-component,

$$\begin{pmatrix} A_x \\ - \\ A_z \end{pmatrix} \times \begin{pmatrix} B_x \\ - \\ B_z \end{pmatrix}$$

And the term simply is $A_z B_x - A_x b_z$ by looping through the top.

2 Unit and Dimensions

Units are dimension are in many ways what give the abstract ideas of physics some ground in the real life. The dimensions of a physical quantity represent what kind of physical quantity is, while the unit is a measure of that physical quantity.

The distinction is appreciable to know, just not always useful.

§2.1 Dimensions

There are seven fundamental quantities which are assigned specific dimensions. It is not necessary to take these as fundamental, in some perfectly fine formulations like the Planck quantities this is not the case. But we'll not be working in these for now.

Quantity	Unit Name	Unit symbol	Dimension
Mass	kilogram	kg	M
Length	meter	m	L
Time	second	s	T
Electric Current	ampere	A	I
Luminous Intensity	candela	cd	J
Temperature	kelvin	K	Θ
Amount of substance	mole	mol	N

Table 2.1: The 7 fundamental quantities.

For any physical quantity A, we use the notation $[A]$ to denote its dimension. For instance,

$$[F] = MLT^{-2}$$

Evaluating the dimensions has a really neat use, it allows us to check our answers, and establish some relations through the use of *dimensional* analysis.

§2.2 Rules regarding dimensions

There are particle constrains that help us utilise dimensions quite a lot. For one, the dimensions on both sides of an equation must be homogenous, i.e the same. It is nonsense to ask if 1 second is equal to 1 kelvin.

Another constraint is that the operations of addition and subtraction can only be performed on quantities of the same dimension, and indeed you cannot add a metre to a kilogram.

Thus, we arrive at a third derived constraint, namely that the functions such as sin, cos, exp, log must all have dimensionless constraints.

This because for instance, let us take the taylor expansion of e^x,

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

We note that if we x had a dimension, for example $[x] = L$, then we would be adding L to L^2, L^3 and so on. But this is not permitted, we cannot add a metre to a square metre or a square metre to a cubic metre and so on. Thus, we have that their arguments must be dimensionless.

§2.3 Dimensional Analysis

Suppose that we know that the rest energy of a particle depends on it mass and the speed of light. If we were asked to find a relation determining its rest energy we could use the idea of dimensional analysis.

We see that,

$$E \propto m, c$$

We could note that the dimesion on both side of the equation must have the same dimesions. Thus we need to combine them in such a manner that we get the dimensions of energy.

Since $[c] = LT^{-1}$, $[m] = M$ and $[E] = ML^{-2}T^{-2}$, we can form a guess and say that,

$$E = mc^2,$$

since the dimensions are same across both quantities. In fact this the correct formula!

In general, for some quantity A dependent on parameters $Q_1, Q_2, \dots, Q_n$,

$$A = Q_1^{\alpha_1} \cdot Q_2^{\alpha_2} \cdot \dots \cdot Q_n^{\alpha_n}$$

where each of $\alpha_1, \alpha_2, \dots, \alpha_n$ are constants. Then we simply use dimensional analysis to compute the result.

There are catches, however, suppose that we were asked to find the kinetic energy of the particle which depends on its mass and velocity. We might guess by dimensional analysis alone that it is mv^2, which is obviously false. What dimensional analysis gives us is that,

$$E \propto mv^2$$

And we must attach a constant for a definite answer,

$$E = kmv^2$$

The constants comes out to be $1/2$, but we can't determine it with dimensional analysis alone.

Let us take a particular example now, the pendulum: its period depends on L, g, and the amplitude θ_0.

We note that the amplitude and angles in general are dimensionless since they're arguments of the trig functions. If we use dimensional analysis, we run into a bit of a problem.

If we use dimensional analysis, we see that we can use any combination of θ_0, since they're all dimensionless. In particular, we can use any function of θ_0.

If we ignore θ_0 since its dimensionless, we get that

$$T \propto \sqrt{\frac{L}{g}}$$

We hence attach a dimensionless group, a function of θ_0 to get the period of a pendulum as $T = f(\theta_0)\sqrt{L/g}$.

In general this can be done for any dimensionless group, for instance the resistance of conductor depends on its resistivity, the cross-sectional area, and its length. Say we know their dimensions and get that,

$$R = \rho\frac{L}{A}$$

We could have also formulated by dimensional analysis, that $R = \rho/\sqrt{A}$, the dimensions are the same. In fact, we could multiply it by any function of L^2/A since it is dimensionless.

Theorem 2.1 (Buckingham Pi Theorem) *Dimensional analysis can't always pin down the form of the answer. If one has N quantities with D independent dimensions, then one can form $N-D$ independent dimensionless quantities. Dimensional analysis can't say how the answer depends on them.*

This is one of the limitations of dimensional analysis.

§2.4 Limiting Cases

Sometimes we can verify our answers using limiting cases.

Suppose that we don't remember if the acceleration of a block on a smooth, non-accelerating wedge is $g\sin\theta$ or $g\cos\theta$.

We can then consider the limiting cases of θ. For instance, if $\theta \to 0$ then the block rests on a horizontal plane and experiences no acceleration, which is only satisfied by $g\sin\theta$. We could have also looked at $\theta \to \pi/2$ as another limiting case.

PART A

Mechanics

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3 Kinematics

Kinematics is the study of motion without concerning its cause. It allows us to calculate and find out the evolution of a body. Everything in our world undergoes motion, one way or the other. To begin our study of motion, let us first concern ourselves with some definitions.

§3.1 Reference Frame and Point Particle

Motion of any object is considered to relative to an observer. The observer defines a particular co-ordinate system called the reference frame

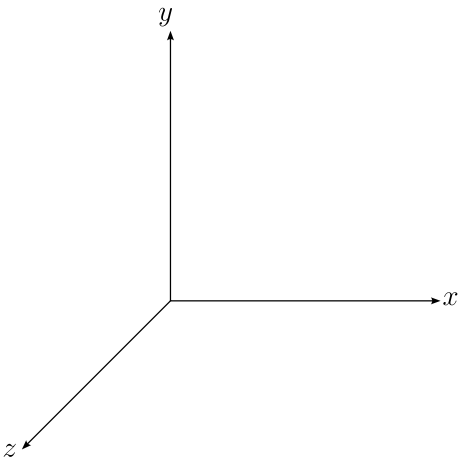


Figure 3.1: A Reference Frame

We use a reference frame according to our convenience. In general, we use the 3-dimensional cartesian system like the one in fig. 3.1.

Definition 3.1 (Reference Frame)

A reference frame is a co-ordinate frame relative to which motion of a particle is considered.

We shall, until we encounter rigid body systems consider the motion of a point particle. Well, why? It is done so to simplify the calculation of our system. The motion of a (rigid)body is discussed later.

Definition 3.2 (Point Particle)

A particle whose size is negligible in the study of its motion is called a point particle.

To discuss the motion, state of our body, it is also necessary to setup some time co-ordinate. It is the reading of the clock in the observer's frame. In general, we setup our clocks at 0 at the start of an event.

§3.2 Position Vector and Displacement

§3.2.1 Position Vector

▶ One thing of importance is that a position vector of a particle is a function of time. We generally denote this as $r(t)$ to show that is a function.

We define the position of a particle relative to a reference frame using a *position vector*. One end of the position vector is the origin of our reference frame. The other is the particle itself. Consider fig. 3.2, the vector $\boldsymbol{r}$ from O to A. It describes the position of A relative to the co-ordinate frame. Let $A = (x, y, z)$. Then we denoted $\boldsymbol{r}$ as $\boldsymbol{r}(x, y, z)$.

The position to any point is written as,

$$\boldsymbol{r} = (x, y, z) = x\hat{\boldsymbol{i}} + y\hat{\boldsymbol{j}} + z\hat{\boldsymbol{k}}$$

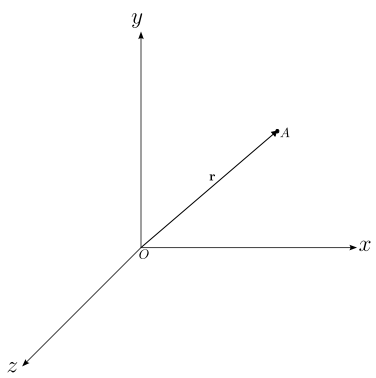


Figure 3.2: A position vector, $\mathbf{r}$.

§3.2.2 Displacement

▶ Note that the position vector is not a *true* vector. It is tied to particular reference frame.

Consider the movement of a particle from $A = (x_1, y_1, z_1)$ to $B = (x_2, y_2, z_2)$. The *displacement* is the change in the position vector, and defines a true vector $\boldsymbol{s} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$. $\boldsymbol{s}$ is called the displacement vector. Note that it contains no information about the individual positions, but only about the relative position of each. Thus, our choice of reference frame, and thus position vector does not matter when we're concerned with displacement. In fig. 3.3, the vector $\boldsymbol{S}_{AB}$ defines a displacement from A to B.

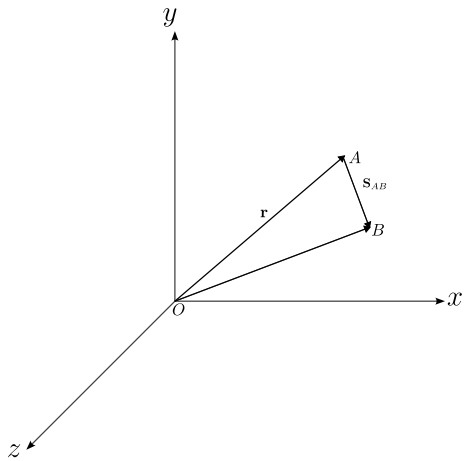


Figure 3.3: Displacement from A to B

The difference between displacement and distance is that of the path they describe. The distance covered is the length of the actual path, while the magnitude of the displacement is simply the length of the vector between the final and initial positions.

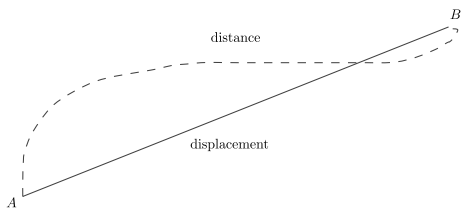


Figure 3.4: Distance

Algorithm 3.3 — Consider the case of displacement and distance of a particle in circular motion with radius r Let it move from A to B, inscribing an angle θ (in radians) Then, we have,

Distance between A, $B = r\theta$

Displacement between A, $B = 2r\sin\left(\frac{\theta}{2}\right)$

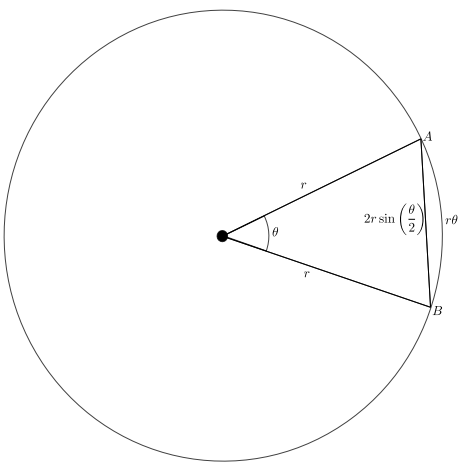


Figure 3.5: Displacement in a circle

§3.3 Speed and Velocity

We are already familiar with the notion of speed. In elementary terms, speed is *distance/time*. Speed itself is an elementary concept. It tells us nothing about the direction of the object whose speed we are talking into consideration. Thus, it is a *scalar* quantity.

The Velocity of a particle, contrastingly, contains information about both the speed and direction of the object in consideration. It is a vector. Similar to

▶ The symbol $\equiv$ stands for 'defined as'.

position, it is also generally a function of time $v(t)$, though it may also be dependent on the position $v(x)$.

Definition 3.4 (Velocity)

Velocity is defined as,

$$\boldsymbol{v} \equiv \dot{\boldsymbol{r}} \tag{3.1}$$

One thing of note is that velocity can also be expressed as the derivative of displacement, which is just $\boldsymbol{r} - \boldsymbol{r}_0$. Since the latter term is constant, we simply get that $\dot{\boldsymbol{s}} = \dot{\boldsymbol{r}}$.

From this definition, we may gather that,

$$\int_{r_0}^{r} d\boldsymbol{r} = \int_0^t \boldsymbol{v}\, dt$$

Or,

$$\boldsymbol{r} = \boldsymbol{r}_0 + \int_0^t \boldsymbol{v}\, dt \tag{3.2}$$

Sometimes, we do not need to compute the derivative of the position vector, and simply need an average estimate, or in the case of uniformly accelerated motion, it is not required that we computer derivatives. In such a case we use the concept of **average velocity**,

$$\bar{\boldsymbol{v}}_{12} = \frac{\Delta \boldsymbol{s}}{t_2 - t_1}$$

▶ The dot over r in $\dot{\boldsymbol{r}}$ tells us that r has been differentiated with respect to time.

Algorithm 3.5 — Let us consider the cases of average velocity,

Case I When there are n equal time intervals with velocity $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n$. Then,

$$\bar{\boldsymbol{v}} = \frac{\boldsymbol{v}_1 + \boldsymbol{v}_2 + \cdots + \boldsymbol{v}_n}{n}$$

Case II When there are n equal intervals of distance/displacement with velocity $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n$. Then,

$$\bar{\boldsymbol{v}} = \frac{n}{\frac{1}{\boldsymbol{v}_1} + \frac{1}{\boldsymbol{v}_2} + \cdots + \frac{1}{\boldsymbol{v}_n}}$$

In general velocity is also referred to as 'instantaneous velocity'. This is done to remind of the difference with average velocity. We will not refer to it that way since by velocity we shall always mean $\boldsymbol{v}$ as defined in definition 3.4. The same goes for speed. Notably, speed is not defined as $\dot{d}$ where d is the distance.

While the average speed is very well that, speed is actually the magnitude of the velocity of that point. Consider the distance between a and b as $a \to b$. The path between them approaches a straight line. Thus, the distance and displacement become the same in the limiting case. Therefore, the *magnitude* of velocity is the speed at that instant.

Therefore,

$$\text{speed} = \left| \frac{d\boldsymbol{r}}{dt} \right|$$

Algorithm 3.6 — Let us consider some techniques for solving equations of velocity. Consider the case when velocity and time are given and we have to figure out displacement or when displacement and and velocity are given, but time is not.

1. $v = c$: This is a trivial case. Just use equations of uniformly accelerated motion.
2. $v = f(x)$: $\dfrac{dx}{dt} = f(x)$. Then, $\dfrac{dx}{f(x)} = dt$. Now, we simply have,
$$\int_{x_1}^{x_2} \frac{dx}{f(x)} = \int_{t_1}^{t_2} dt$$
to get x_1 and x_2 or t_1 and t_2. We may also set our co-ordinates such that $x_2 = x$, $x_1 = 0$ and $t_2 = t$, $t_1 = 0$.
3. $v = f(t)$: This is also easy, just, $f(x) = \dfrac{dx}{dt}$, then,
$$\int_{t_1}^{t_2} f(t)\, dt = \int_{x_1}^{x_2} dx$$
We simply get x_1 and x_2 or t_1 and t_2 by solving the integral.

§3.4 Acceleration

If a body does not move with uniform velocity, it accelerates. Acceleration is also a vector and is the rate of change of velocity.

Definition 3.7 (Acceleration)

Acceleration is defined as,
$$\boldsymbol{a} \equiv \dot{\boldsymbol{v}} = \ddot{\boldsymbol{r}} \tag{3.3}$$

Much like the velocity definition, we have,

$$\boldsymbol{v} = \boldsymbol{v}_0 + \int_0^t \boldsymbol{a}\, dt \tag{3.4}$$

There is also another way to represent acceleration, as,

$$\boldsymbol{a} = \boldsymbol{v}\frac{d\boldsymbol{v}}{dx}$$

Such a representation is particularly useful when the velocity of a particle is a function of its position. In such a case, derivating with time is quite a hassle.

§3.5 Uniformly Accelerated Motion

When the acceleration of a particle, a is constant, it undergoes uniformly accelerated motion. We use special cases of integrals of a and v with respect to time and position for deriving the equations of importance.

Theorem 3.8 *For a particle with initial velocity u undergoing uniformly accelerated motion of acceleration a,*

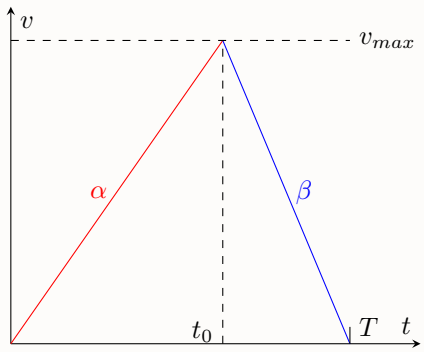
$$v = u + at \tag{3.5}$$
$$v^2 = u^2 + 2as \tag{3.6}$$
$$s = ut + \frac{1}{2}at^2 \tag{3.7}$$
$$s_n = u + \frac{a}{2}(2n - 1) \tag{3.8}$$

The derivation of these is quite easy, integrate $\int a\,dx$ for the first, $\int a\,dt$ for the second (both with constant acceleration), substitute from the former into the latter for the third, and finally for n^{th} second just do $s_n - s_{n-1}$.

Now, we will be looking at some fascinating questions that I have encountered.

Example 3.9 Consider a particle moving with acceleration α initially. After some time, it starts to 'de-accelerate' at β acceleration. After time $t = T$, the particle comes to a stop. Find the displacement covered by particle in time T.

Solution. It might be tempting to start calculating after setting up time $t = t_0$ when acceleration changed from α to β, i.e., the particle moves with acceleration α till $t = t_0$.
However, let us try to do a cleaner solution. Consider the $v - t$ graph of the particle,



Note that $\alpha(t_0) = \beta(T - t_0) = v_{max}$. Thus,

$$\begin{aligned}
\alpha(t_0) &= \beta(T - t_0) \\
\alpha(t_0) &= \beta(T) - \beta(t_0) \\
\alpha(t_0) + \beta(t_0) &= \beta(T) \\
t_0 &= \frac{\beta T}{\alpha + \beta} \\
v_{max} &= \alpha t_0 \\
v_{max} &= \frac{\alpha\beta T}{\alpha + \beta}
\end{aligned}$$

Now, note that $\frac{1}{2} \times v_{max} \times T$ is the area under the graph.

The area under the $v - t$ is the total displacement.

Thus, displacement, $s = \dfrac{\alpha\beta T^2}{2(\alpha+\beta)}$. □

Example 3.10 A car is at distance d from a boy. It starts accelerating at a m/s^2. What is the minimum velocity that the boy should have to catch up with the car.

Solution. Consider separation of boy and car, $s_{c,b}$. Using the equations of motion, we have $s_b = vt$, $s_c = \frac{1}{2}at^2$. Thus,

$$s_{c,b} = d + \frac{1}{2}at^2 - vt \tag{3.9}$$

The most important thing to note here is that,

For two objects to meet, the solution for time must be real.

Therefore, from the simple equation, (3.9), we must have, for real solutions, $b^2 - 4ac$ for the equation $at^2 + bt + c$. Solving for this by substituting values from (3.9), we have,

$$v \geqslant \sqrt{2ad} \tag{3.10}$$

□

Remark 3.11. One thing of note is that the converse of above is true as well,

For two objects to *not* meet, the solution of time must be imaginary or negative.

One more key note to takeaway is that,

At instant of maximum separation, relative velocity of particles is 0.

Example 3.12 A body is dropped at $t = 0$, after time $t = t_0$, another body is thrown downwards with velocity u m/s. Assuming first body reaches ground first, plot graph of separation.

Solution. At instant t_0, displacement of first particle $= \frac{1}{2}gt_0^2$. Note that here we set up co-ordinates such that positive y is downwards from point of drop. The displacement of first body at $t = t$ after t_0 but before $\boxed{\text{reaching ground}}$ is

$$s_1 = \frac{1}{2}gt_0^2 + \frac{1}{2}g(t - t_0)^2 \tag{3.11}$$

While for second body is,

$$s_2 = ut + \frac{1}{2}g(t - t_0)^2 \tag{3.12}$$

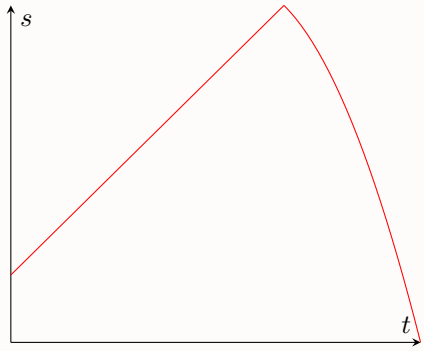
Thus,

$$s_{1,2} = \frac{1}{2}g(t_0)^2 + t(gt_0 - u) \tag{3.13}$$

Thus, before reaching ground, $s - t$ graph is linear. However, after first body reaches ground,

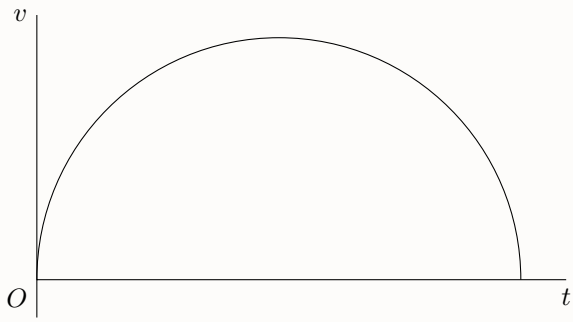
$$s_{1,2} = ut + \frac{1}{2}g(t)^2 \tag{3.14}$$

which is a parabola. Thus, overall graph is,



□

Example 3.13 Find the total displacement if the $v - t$ graph is as follows,



The idea behind this is simple, the graph here is **not** a semicircle. It is an ellipse, because of dimensional limitations! The equations of circle, $v^2 + t^2 = r^2$ doesn't work because we cannot add two dimensionally different quantities.

Therefore, the graph is an ellipse, and we conclude by computing the area of the ellipse. I haven't given the data, since I wanted to mention the idea.

§3.6 Motion in Polar Co-ordinates

The polar co-ordinate axes are the two-dimesional subset of the 3-d cylindrical co-ordinates. Any point in polar co-ordinates is depicted by a system of two unit vectors, $\hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{\theta}}$. However, each of these are $\boxed{\text{dependent on the position of the particle}}$ and maybe written explicitly as $\hat{\boldsymbol{r}}(\theta)$ and $\hat{\boldsymbol{\theta}}(\theta)$.

Here, $\hat{\boldsymbol{r}}$ points in the direction of increasing radius(along the radial vector) $\hat{\boldsymbol{\theta}}$ points in the direction of increasing θ (tangent to the radial vector). These two unit vectors are *orthogonal* at any point.

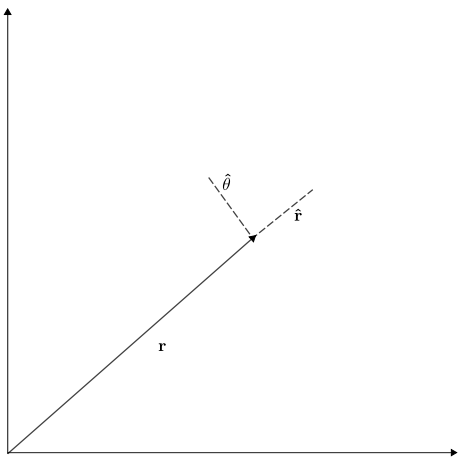


Figure 3.6: Unit vectors in polar co-ordinates

In fig. 3.6 if $\boldsymbol{r}$ makes an angle θ with the horizontal, then,

$$\hat{\boldsymbol{r}} = \cos\theta\hat{\boldsymbol{i}} + \sin\theta\hat{\boldsymbol{j}} \tag{3.15}$$

$$\hat{\boldsymbol{\theta}} = -\sin\theta\hat{\boldsymbol{i}} + \cos\theta\hat{\boldsymbol{j}} \tag{3.16}$$

§3.7 Kinematics in Polar Co-ordinates

Now let us formulate kinematics in polar co-ordinates. The position $\boldsymbol{r}$ can be written as

$$\boldsymbol{r} = r\hat{\boldsymbol{r}} \tag{3.17}$$

Velocity follows normally,

$$\boldsymbol{v} = \dot{\boldsymbol{r}} = \dot{r}\hat{\boldsymbol{r}} + r\frac{d\hat{\boldsymbol{r}}}{dt} \tag{3.18}$$

or does it? What even is $\frac{d\hat{\boldsymbol{r}}}{dt}$? Well the answer lies in the time-derivative of the unit vectors. Using the definitions in (3.15), we get,

$$\frac{d\hat{\boldsymbol{r}}}{dt} = \dot{\theta}\hat{\boldsymbol{\theta}} \tag{3.19}$$

$$\frac{d\hat{\boldsymbol{\theta}}}{dt} = -\dot{\theta}\hat{\boldsymbol{r}} \tag{3.20}$$

Using these results, we get that

$$\boxed{\begin{aligned} \boldsymbol{v} &= \dot{r}\hat{\boldsymbol{r}} + r\dot{\theta}\hat{\boldsymbol{\theta}} && (3.21)\\ \boldsymbol{a} &= (\ddot{r} - r\dot{\theta}^2)\hat{\boldsymbol{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\boldsymbol{\theta}} && (3.22)\end{aligned}}$$

The most interesting result are the various terms of acceleration in (3.22). First let us consider the terms along $\hat{\boldsymbol{r}}$. $\ddot{r}$ is the *radial* acceleration. It is the change is radial speed. The second term $-r\dot{\theta}^2$ is the *centripetal* acceleration. It arises from a change in tangential velocity.

Now let us consider the terms along $\hat{\boldsymbol{\theta}}$. The $r\ddot{\theta}$ term is the tangential acceleration resulting from change in tangential speed. The other term $2\dot{r}\dot{\theta}$ is the *coriolis* acceleration. I shall discuss it when discussing rotating frames.

§3.8 Free Fall

When an object is dropped downwards freely, it undergoes *free fall*. We ignore air resistance, until we're told not to and in general also talk about cases of the object being thrown instead of dropped. An object dropped from any height undergoes accelerations $\boldsymbol{a}$ where $\|\boldsymbol{a}\| = g$. The direction and sign of acceleration is entirely dependent on us.

In general, it is a good idea to setup our axes such that the direction above ground is positive and towards the ground is negative. The acceleration during free fall is always towards the ground. Therefore, for these cases $\boldsymbol{a} = -g\hat{\boldsymbol{j}}$.

Some important points of note are that "g" does not become positive or negative or whatever, this is a common misconception. Instead, we just decide where exactly the positive direction is and setup g according to that. When a ball is thrown with some velocity u say in the positive directions (change it into $-u$ if in the negative direction), then we have,

Algorithm 3.14 — The following are special cases of equations of motion.

1. $v = u - gt$
2. $v^2 = u^2 - 2gh$
3. $h = ut - \frac{1}{2}gt^2$

The case when a ball is thrown upwards is exactly symmetrical to this one, but there the ball goes up, reachs $v = 0$ and then goes down. Also note that all of these equations are vector equations.

Example 3.15 Two balls are thrown in intervals of 2 seconds with velocity u each. Find the time when they collide.

Solution. Note that when they collide, their displacement from ground must be the same! Setting the origin at ground, and direction of away from ground as positive, we have,

For the first ball, $h_1 = ut - \frac{1}{2}gt^2$. Since the second ball is thrown 2 seconds later, $h_2 = u(t-2) - \frac{1}{2}g(t-2)^2$. Now, we just have $h_1 = h_2$ and the calculation is trivial. □

The thing to note in this example is that you do not need to setup two cases, one in which g is negative and one in which g is positive since it is in the direction of velocity. Just note that these equations are vectorial and we are done.

Some quick formulas to remember because they help a lot,

Algorithm 3.16 — For an object dropped from height h,

1. Time to reach ground, $t = \sqrt{\frac{2h}{g}}$.
2. Velocity when it reaches ground $v = \sqrt{2gh}$.

Deriving them is easy, and they can be generalised as when it travels height h between the point of drop to anywhere, instead of ground. Clearly, both still follow there.

Also, always remember,

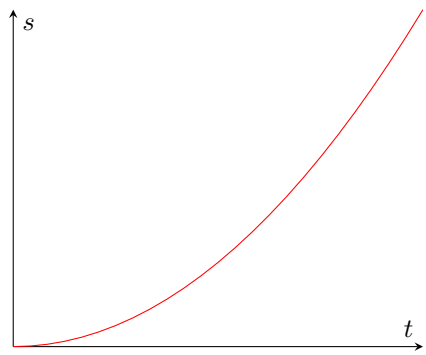
When an object is thrown/dropped from another object traveling with some velocity and acceleration, it inherits its velocity but not its acceleration.

add screw/lift example

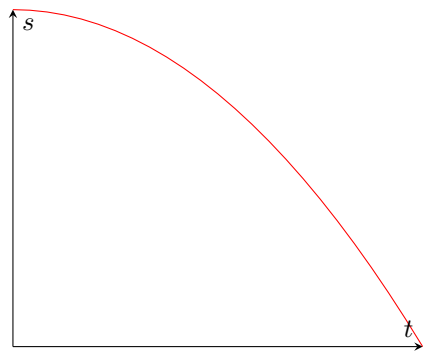
§3.9 Graphs in Motion

In the graphs we are going to discuss, acceleration will either be 0 or constant. These graphs help to quickly solve problems, especially related to total distance and displacement.

Let us consider the direction away from origin positive, and towards origin negative. Then, consider constant acceleration, $a = c$. We have the following displacement time graphs,



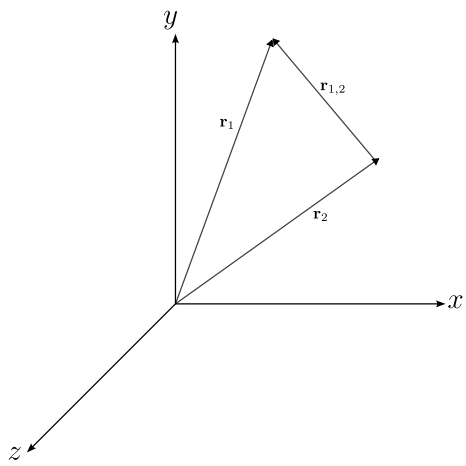
(a) For $c > 0$.



(b) For $c < 0$

§3.10 Relative Motion

There is a very neat technique to solve questions of kinematics, it is to switch our reference frames. Let the base reference frame be that of earth, then,



If $\boldsymbol{r}_1$ evolves with time, $r_1(t)$. And so does $r_2(t)$. Then,

$$\begin{aligned}\boldsymbol{r}_1 &= \boldsymbol{r}_{1,2} + \boldsymbol{r}_2\\ \dot{\boldsymbol{r}}_1 &= \dot{\boldsymbol{r}}_{1,2} + \dot{\boldsymbol{r}}_2\\ \boldsymbol{v}_1 &= \boldsymbol{v}_{1,2} + \boldsymbol{v}_2\\ \boldsymbol{v}_{1,2} &= \boldsymbol{v}_1 - \boldsymbol{v}_2\end{aligned}$$

Obviously if the vectors face the same direction in the same plane, then, the components subtract. If they face opposite direction, then the components add.

$\boldsymbol{v}_{1,2}$ is called the velocity of *1* w.r.t *2*. When doing problems, try to setup a reference frame such that velocity of any single object vanishes. This allows for a very neat simplification of problems.

§3.11 Motion in higher dimensions

We can mostly safely extend our results from one dimension to higher dimensions. So for instances, velocity of a particle whose trajectory is defined as $x\hat{\boldsymbol{\imath}}+y\hat{\boldsymbol{\jmath}}+z\hat{\boldsymbol{k}}$ can simply be computed as,

$$v = \dot{x}\hat{\boldsymbol{\imath}} + \dot{y}\hat{\boldsymbol{\jmath}} + \dot{z}\hat{\boldsymbol{k}}$$

Such a thing is not true for non-cartesian co-ordinates, as we say in the case of polar forms. Due to the orthogonality of the cartesian unit vectors, and using dot products we simply have,

$$v_x = \dot{x}, v_y = \dot{y}, v_z = \dot{z}$$

And the same is true for all other vectors. This gives us the power to resolve our vector components into components alongside unit vectors and consider independent one dimensional motion for each of them.

§3.12 Projectile Motion

Projectile motion is one of the most important motion to consider in 2 dimensions. It is the motion of a projected body. Generally, we only care about the motion of a body projected from ground under the influence of gravity.

Let a body be projected at an angle θ from the ground at velocity $\boldsymbol{v}$ along θ. Then, by resolving $\boldsymbol{v}$ into $v_x\hat{\boldsymbol{\imath}} + v_y\hat{\boldsymbol{\jmath}}$, we have,

$$\begin{aligned}x &= v_x t\hat{\boldsymbol{\imath}}\\ y &= v_y t - \frac{1}{2}gt^2\hat{\boldsymbol{\jmath}}\end{aligned}$$

Doing some manipulations, we may note that $v_x = v\sin(\theta)$, $v_y = v\cos(\theta)$. Also, substituting $t = x/v\sin(\theta)$, we have,

$$\boxed{y = x\tan(\theta) - \frac{gx^2}{2v^2\cos^2(\theta)}} \tag{3.23}$$

The path of the projectile is parabolic as shown in fig. 3.8.

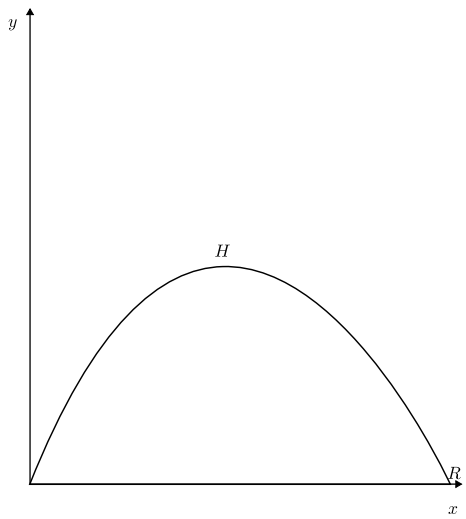


Figure 3.8: Trajectory of a projectile

Here, H is the maximum height of the particle. Clearly, when the projectile is at H, $v_y = 0$. We have, $v_y^2 = 2gH$,

$$\boxed{H = \frac{u^2 \sin^2(\theta)}{2g}} \tag{3.24}$$

The time period of the projectile can be simply calculated as well, since at the final time, T, $y = v_y t - 1/2gt^2 = 0$ and,

$$\boxed{T = \frac{2v \sin(\theta)}{g}} \tag{3.25}$$

R is the range the particle covers horizontally in time T and is clearly,

$$\boxed{R = \frac{v^2 \sin(2\theta)}{g}} \tag{3.26}$$

Note that if the particle is projected at $\theta = \pi/4$, the range is maximum and is $u^2/2g$. Otherwise, for $0 < \theta < \pi/2$, $\sin(2\theta)$ attains the same value for two θ_1, θ_2. These θ_1, θ_2 are complementary by trigonometry.

More generally, if we consider the projectile to have an initial position of (x_0, y_0), then, by substituting for time, we find that the vertex of the parabola is $\left(y_0 + \frac{u^2 \sin^2 \theta}{2g}, x_0 + \frac{u^2 \sin 2\theta}{g}\right)$.

The range is a little more difficult to computer here, as the particle will transverse to the ground which is lower than y_0. Freely setting up our co-ordinates as $(h, 0)$, we get by the equation of trajectory,

$$0 = h + R\tan(\theta) - \frac{gx^2}{2v^2 \cos^2(\theta)}$$

Derivating this with respect to theta to find the maxima, and then using the attained value, we have,

$$R = \frac{u}{g}\sqrt{u^2 + 2gh} \tag{3.27}$$

Some peculiar things about projectile motion is that, a) it is reversible, and b) the time to attain some vertical displacement can be found in terms of the maximum height alone.

Consider the vertical position of the particle to be s at some time t. Then,

$$s = u_y t - \frac{1}{2}gt^2$$

Also, note that if the maximum height is h,

$$u_y = \pm\sqrt{2gh}$$

Where the positive value is during the ascent and the negative is during the descent. Then,

$$s = \pm\sqrt{2gh}t - \frac{1}{2}gt^2$$

Solving for t gives us,

$$t = \pm\sqrt{\frac{2h}{g}} \pm \sqrt{\frac{2(h-s)}{g}}$$

If the initial velocity is positive and $s > 0$,

$$t = \sqrt{\frac{2h}{g}} \pm \sqrt{\frac{2(h-s)}{g}} \tag{3.28}$$

the two values correspond to the fact that the particle will have the same vertical displacement at two instances of time.

If the initial velocity is positive but $s < 0$, i.e the particle starts at some height but then transverses below it,

$$t = \sqrt{\frac{2h}{g}} + \sqrt{\frac{2(h-s)}{g}} \tag{3.29}$$

is the only reasonable solution.

If the initial velocity is negative, then clearly the particle will attain a lower position. The maximum height in this case is simply the initial height of the particle.

$$t = -\sqrt{\frac{2h}{g}} + \sqrt{\frac{2(h-s)}{g}} \tag{3.30}$$

add drag forces

§3.13 Drag Forces

In a more realistic scenario, a particle undergoing motion in any media experiences a drag force, anti-parallel to its velocity. In particular,

$$\boldsymbol{a} = -\alpha v\hat{\boldsymbol{v}} - \beta v^2\hat{\boldsymbol{v}}$$

where the linear term arises from a property of the media, the viscosity(we will talk about it in the later sections.)

The quadratic term is a result of the collision of the particles with atoms and molecules during its motion. In general, the quadratic term dominates for higher velocities.

It is very difficult to analyse motion with drag, but we shall have a look at some particular cases of linear drag.

§4.2.2 Newton's Second Law

Example 3.17 Consider a projectile of mass m thrown upwards with initial velocity v_0. It undergoes a linear drag force, $\mathbf{F} = -kv\hat{\boldsymbol{v}}$. Find the velocity of the particle as a function of time.

Solution. Using newton's second law, also by noting that it also undergoes gravitation, $\boldsymbol{a} = -g - kv/m\hat{\boldsymbol{v}}$.

Now, we note that,

$$\frac{dv}{dt} = -g - \frac{kv}{m}$$

$$\frac{dv}{g + kv/m} = -\,dt$$

$$\int_{v_0}^{v_f} \frac{dv}{g + kv/m} = \int_0^t dt$$

$$\frac{m}{k}\log(g + kv/m)\Big|_{v_0}^{v_f} = t$$

Which finally gives us,

$$v(t) = e^{-kt/m}v_0 + \frac{gm}{k}(e^{-kt/m} - 1) \tag{3.31}$$

□

§3.14 Kinematics of a rigid body

When we describe the motion of a point particle, we need 3 co-ordinates to do so in free space. For a system of n particles, we thus need $3n$ *generalised co-ordinates*. This makes a description of motion rather difficult.

However, when we concern ourselves with rigid bodies, the problem becomes much easier. A rigid body is a system of particles such that they are always at a fixed distance apart from each other. For instance, the end points of a rod.

Rigid bodies, like everything in physics are a good approximation. The distances won't always be equal (for instance, at the time of collision). But it is enough of an approximation that we'll let it go.

By Chasles' theorem, which we'll only state, we only 6 co-ordinates to describe a rigid body. Generally, we use 3 translational and 3 angular ones.

Here, we're only going to look into the rotation of a rigid body.

Take an arbitrary point P on the body and set up cylindrical co-ordinates around it. Which are polar co-ordinates with the z axis. We'll also set up an $x - y$ plane. We note that the body rotates about P in such a frame.

In cylindrical coordinates, every point on the body can be ascribed a coordinate θ, the angle between the projection of its position vector on the $x - y$ plane. This is called the azimuthal angle.

The other two co-ordinates are r, the distance of a particle from the z axis, and z the signed height of the body.

If a particle on a rigid body undergoes a change in its azimuth by $d\theta$, then because of the constraints of a rigid body, every particle undergoes a change of $d\theta$. Thus, we define the angular speed of the rigid body, ω to be $d\theta/\,dt$.

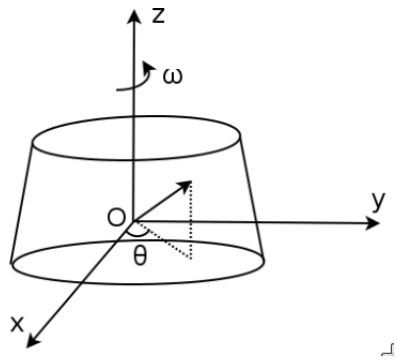


Figure 3.9: Rigid Body about P

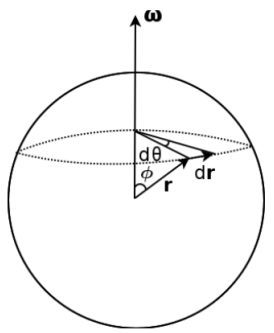


Figure 3.10: $\boldsymbol{\omega}$ and $d\boldsymbol{r}$.

In the case of rigid body motion, we also define $\boldsymbol{\omega}$, the angular velocity. It has the magnitude ω but points in the direction perpendicular to plane of rotation. We define the unique direction by the right hand grip rule. If you curl your fingers in the direction of rotation, then the direction in which your thumb points is the direction of $\boldsymbol{\omega}$.

Consider a particle at a position $\boldsymbol{r}$ from P, and makes an angle ϕ with $\boldsymbol{\omega}$.

Then, if it undergoes a change of $d\theta$ azimuthally, it moves through a circle of radius $r\sin\phi$ and transverses a distance $\|d\boldsymbol{r}\| = r\sin\phi\, d\theta = r\sin\phi\,\omega\, dt$.

Thus,

$$d\boldsymbol{r} = \boldsymbol{\omega} \times \boldsymbol{r}\, dt,$$

And,

$$\boxed{\frac{d\boldsymbol{r}}{dt} = \boldsymbol{v} = \boldsymbol{\omega} \times \boldsymbol{r}} \tag{3.32}$$

One thing we note across here is that if we take any another point on the rigid body, say Q which has a velocity $\boldsymbol{v}_Q$ in the frame of P, then,

$$\boldsymbol{v}_Q = \boldsymbol{\omega} \times \boldsymbol{r}_{PQ}$$

In the frame of Q, let us suppose that the body has an angular velocity, $\boldsymbol{\omega}_1$. Then the velocity of P in the frame of Q must be

$$\boldsymbol{v}_P = \boldsymbol{\omega}_1 \times \boldsymbol{r}_{QP} = -\boldsymbol{\omega}_1 \times \boldsymbol{r}_{PQ}$$

Since $\boldsymbol{v}_P = -\boldsymbol{v}_Q$ if Q is non-rotating,

$$\boldsymbol{\omega}_1 \times \boldsymbol{r}_{PQ} = \boldsymbol{\omega} \times \boldsymbol{r}_{PQ}$$

which gives us that $\boldsymbol{\omega}_1 = \boldsymbol{\omega}$. This means that in any non-rotating frame with respect to P, the angular velocity is the same.

> ▶
> $\boldsymbol{\omega}$ is defined to be positive along the positive z axis.

Basically, the angular velocity of a rigid body is intrinsic. It does not depend upon the arbitrary point on the rigid body about which we establish our co-ordinates.

This is in contrast to something like the velocity which does depend on it.

Generally, If the particle P has a velocity $\boldsymbol{v}_{ref}$ in the lab frame, then the velocity of a point at a position $\boldsymbol{r}$ from P in the lab frame is,

$$\boldsymbol{v} = \boldsymbol{v}_{ref} + \boldsymbol{\omega} \times \boldsymbol{r}. \tag{3.33}$$

§3.14.1 Instantaneous center of Rotation

Equation (3.33) can be seemingly simplified by using a point known as the Instantaneous Center of Rotation, ICoR.

Definition 3.18

The Instantaneous Center of Rotation is defined as the point that is fixed to the frame of the rigid body — but not necessarily lying on the body — that has zero velocity in the lab frame at that particular instant.

The concept of being fixed to the "frame" of the body is a weird one. You may imagine it as something like a plane being attached to the body. This planes rotates with the body. Now the ICoR is a point on the plane that has a 0 velocity in the lab frame.

It does not, necessarily, lie on the body, but on the "frame" of the body, where it rotates just as the body does. Of course, this talk to an attached plane is merely hypothetical. The ICoR lies on the extension of the body.

We may note that since the ICoR, which we shall denoted I, has zero velocity in the lab frame, the velocity of any point P is simply,

$$\boldsymbol{v}_P = \boldsymbol{\omega} \times \boldsymbol{r}_{IP}$$

Thus the velocity at the point P is perpendicular to the line joining the ICoR and P.

For parallel velocities, thus, we may see that the ICoR lies on the line joining them.

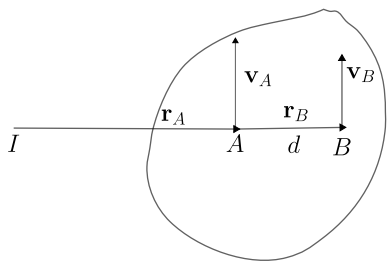


Figure 3.11: ICoR for parallel velocities

Here,

$$v_A = \omega r_A,\ v_B = \omega r_B,$$

And,

$$\frac{v_A}{r_A} = \frac{v_B}{r_A + d}$$

This allows us to greatly simplify and find for not only ICoR but the velocities from some type of argument to figure out the distances.

However, it is a bit more complicated for non-parallel velocities.

Consider fig. 3.12. The ICoR lies on the intersection point of the lines perpendicular to the velocity. Unfortunately, there is general way to calculate the ICoR and some geometric argument must be considered.

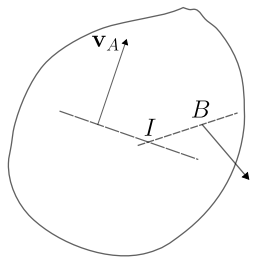


Figure 3.12: ICoR for non-parallel velocities

§3.14.2 Additive Property of Angular Velocities

Angular velocities, much like velocities, are additive across reference frames, i.e, for a rigid body B,

$$\boxed{\boldsymbol{\omega}_{B1} = \boldsymbol{\omega}_{B2} + \boldsymbol{\omega}_{21}} \tag{3.34}$$

Let us sketch out a proof of this. First, let us mention one extremely important equation that will be of much use,

If a particle has velocities $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ in two reference frames, the axes of frame 2 are rotating at $\boldsymbol{\omega}_{21}$ with respect to the first frame and $\boldsymbol{r}_1$ is the position vector of the particle in the first frame,

$$\boxed{\boldsymbol{v}_1 = \boldsymbol{v}_2 + \boldsymbol{\omega}_{21} \times \boldsymbol{r}} \tag{3.35}$$

While eq. (3.35) bears a particular similarity to eq. (3.33), they are very different.

Equation (3.35) tells us the velocity of a particle across two **rotating** frames, while eq. (3.33) tells us the velocity of particle relative to the lab frame by using a reference point on the rigid body.

We won't prove this equation now but rather use it to prove eq. (3.34).

Proof. Let R be the ICoR in the first reference frame. Then, for a rigid body B and the point on the body, P,

$$\boldsymbol{v}_{P1} = \boldsymbol{\omega}_{B1} \times \boldsymbol{r}_{RP}.$$

The velocity of P in frame 2 is,

$$\boldsymbol{v}_{P2} = \boldsymbol{v}_{R2} + \boldsymbol{\omega}_{B2} \times \boldsymbol{r}_{RP}.$$

Now, using eq. (3.35) for $\boldsymbol{v}R1$, $\boldsymbol{v}R2$,

$$\boldsymbol{v}_{R1} = \boldsymbol{v}_{R2} + \boldsymbol{\omega}_{21} \times \boldsymbol{r}_{R1},$$

where $\boldsymbol{v}_{R1} = 0$ because it is the ICoR. This gives us,

$$\boldsymbol{v}_{P2} = \boldsymbol{\omega}_{B2} \times \boldsymbol{r}_{RP} - \boldsymbol{\omega}_{21} \times \boldsymbol{r}_{R1}$$

Finally, applying eq. (3.35) for $\boldsymbol{v}_{P1}$ and $\boldsymbol{v}_{P2}$,

$$\boldsymbol{v}_{P1} = \boldsymbol{v}_{P2} + \boldsymbol{\omega}_{21} \times \boldsymbol{r}_{P1},$$

And,

$$\begin{aligned}
\boldsymbol{\omega}_{B1} \times \boldsymbol{r}_{RP} &= \boldsymbol{\omega}_{B2} \times \boldsymbol{r}_{RP} - \boldsymbol{\omega}_{21} \times \boldsymbol{r}_{R1} + \boldsymbol{\omega}_{21} \times \boldsymbol{r}_{P1} \\
\boldsymbol{\omega}_{B1} \times \boldsymbol{r}_{RP} &= \boldsymbol{\omega}_{B2} \times \boldsymbol{r}_{RP} + \boldsymbol{\omega}_{21} \times (\boldsymbol{r}_{R1} - \boldsymbol{r}_{P1}) \\
\boldsymbol{\omega}_{B1} \times \boldsymbol{r}_{RP} &= \boldsymbol{\omega}_{B2} \times \boldsymbol{r}_{RP} + \boldsymbol{\omega}_{21} \times \boldsymbol{r}_{RP}
\end{aligned}$$

Since P was an arbitrary point, we can safely remove it,

$$\boldsymbol{\omega}_{B1} = \boldsymbol{\omega}_{B2} + \boldsymbol{\omega}_{21}.$$

□

§3.15 Angular Acceleration

Definition 3.19

The angular acceleration of a rigid body is the rate of change of its angular velocity,

$$\boldsymbol{\alpha} = \dot{\boldsymbol{\omega}} \tag{3.36}$$

Writing it more explicitly,

$$\boldsymbol{\alpha} = \frac{d\omega\hat{\boldsymbol{\omega}}}{dt} = \dot{\omega}\hat{\boldsymbol{\omega}} + \omega\frac{d\hat{\boldsymbol{\omega}}}{dt}$$

For our purposes, we shall only consider fixed axis rotation. That is, rotation in which the direction of angular velocity does not change.

Thus,

$$\boldsymbol{\alpha} = \dot{\omega}\hat{\boldsymbol{\omega}} = \alpha\hat{\boldsymbol{\omega}}$$

The acceleration of an arbitrary point on the rigid body can be computed with respect to the lab frame with the help of another reference point on the body can be computed using eq. (3.33).

$$\frac{d\boldsymbol{v}}{dt} = \dot{\boldsymbol{v}}_{ref} + \frac{d\boldsymbol{\omega}}{dt} \times \boldsymbol{r} + \boldsymbol{\omega} \times \frac{d\boldsymbol{r}}{dt}$$

Here, $\boldsymbol{r}$ is the position of the vector of the body relative to the reference point. Thus, it changes as $\dot{\boldsymbol{r}} = \boldsymbol{v} - \boldsymbol{v}_{ref} = \boldsymbol{\omega} \times \boldsymbol{r}$.

$$\boldsymbol{a} = \boldsymbol{a}_{ref} + \boldsymbol{\alpha} \times \boldsymbol{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r}) \tag{3.37}$$

This equation is general and valid for non-fixed axis rotation as well.

For fixed axis rotation,

$$\boldsymbol{a} = \boldsymbol{a}_{ref} + \alpha\hat{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r}) \tag{3.38}$$

If the rotation and translation of the body is strictly in a 2 dimensional plane, the equation can be greatly simplified as $\hat{\boldsymbol{\omega}} \times \boldsymbol{r} = r\hat{\boldsymbol{\theta}}$ (because $\hat{\boldsymbol{\theta}}$ is along the direction of the cross product as $\boldsymbol{r}$ and $\boldsymbol{\omega}$ are perpendicular).

$\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r})$ can be simplified using the BAC-CAB rule, where,

$$\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r}) = \boldsymbol{\omega}(\boldsymbol{\omega} \cdot \boldsymbol{r}) - \boldsymbol{r}(\boldsymbol{\omega} \cdot \boldsymbol{\omega})$$
$$\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r}) = -\omega^2\boldsymbol{r} = -\omega^2 r\hat{\boldsymbol{r}}$$

Finally, we get,

$$\boldsymbol{a} - \boldsymbol{a}_{ref} = \alpha r\hat{\boldsymbol{\theta}} - \omega^2 r\hat{\boldsymbol{r}} \tag{3.39}$$

Just a point to remember is that we are taking α as signed here, thus remember to use $-\alpha$ if $\alpha = -\alpha\hat{\boldsymbol{\omega}}$ in the equation.

4 Netwon's Laws

In this chapter we shall look at newton's laws of motion. These form the basis of dynamics which is the study of a motion's cause. In general, we solve mechanical problems by first considering its dynamics and then reduce it to a purely kinematical problem.

First let us concern ourselves with the concept of linear momentum.

§4.1 Linear Momentum

Definition 4.1

The Linear Momentum, $\boldsymbol{p}$ of a particle of mass m moving with velocity $\boldsymbol{v}$ is defined as,

$$\boldsymbol{p} = m\boldsymbol{v}$$

The total momentum of a system of particles is simply the sum of momentum of each particle. Momentum is a useful quantity because it is conserved, we'll look at that latter. But for now, it has certain unique properties that make the concept of momentum extremely important.

§4.2 Newton's Three Laws

§4.2.1 Newton's First law

Axiom 1. Objects tend to continue in a state of constant velocity unless acted upon by a net external force.

The first law actually is much better stated as the assertion that inertial frames exist.

It defines inertial frames as such reference frames where objects move at constant velocity without the action of an external force. It also makes an empirical observation that such frames exist. Thus, its part definition and part experimental fact.

In fact an inertial frame may also be stated as a frame which has no acceleration. Here, the fact that we can say that it has no acceleration refers to the newtonian concept that acceleration is absolute. That in any frame of reference, we may know that a body is accelerating, even when velocity is relative and an object may be at rest in certain frames and moving in others.

We may measure the acceleration in any reference frame by placing an accelerometer and the acceleration must come out the same no matter what frame we choose.

Since the inertial frames are non-accelerating, they travel with some constant velocity relative to each other. Thus, a frame that travels with constant velocity to an inertial frame is also a reference frame.

Another way to talk about inertial frames is that they are the frames in which newton's laws hold. Certainly, if our frame is non-inertial, none of the laws will hold in this frame.

§4.2.2 Newton's Second Law

A very nice reference which discusses interaction in detail is [KK14]

Axiom 2. The net external force on a body is proportional to its change in momentum.

Thus, our claim is that

$$\sum \mathbf{F} = \frac{d(m\boldsymbol{v})}{dt} = m\boldsymbol{a},$$

if the mass is constant.

Let us take an interlude here and talk about what mass and force really are.

Mass

Assume that there are two bodies, of masses m_1 and m_2. If they say, undergo an "equal amount" of physical interaction, for instance two blocks are accelerated with the same amount of "push", and are accelerated with accelerations a_1 and a_2, then,

$$m_2 \equiv m_1 \frac{a_1}{a_2}.$$

We define the mass of the first body relative to the other. The accelerations can be absolutely measured using an accelerometer. In such a manner we define the acceleration of any ith body relative to the first one as,

$$m_i \equiv m_1 \frac{a_1}{a_i}$$

We define some standard mass as 1 unit and then define the other masses relative to this. Such a definition is an operational definition.

It is not always useful, and is not theoretical but it does give us a way to measure the mass of any body.

The talk about equal physical interaction is vague and purely experimental concept. It is entirely physical and has no theoretical basis on which we can rely. It may seem a little wooly, but it works.

Force

We say a force is produced by a physical interaction. We dont talk about what forces "are". Merely, how they come to be. Whenever we interact physically with an object, we apply a force on it.

It might be tempting to ask in case of forces where such an interaction is not visible, say gravity, does this really hold? In fact, it does the interaction is mediated through the use of fields. We'll have a look at them later.

§4.2.3 Newton's Third Law

The final law describes how forces come in pairs and result of a physical interaction.

Axiom 3. If a body A applies a force on a body B, then the body B applies an equal and opposite force on A. These forces are called action-reaction pairs.

That is to say,

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA}.$$

The third law is also known as the weak law of action and reaction — in direct contrast with the strong law which requires the equal and opposite forces to lie on the same line of action. Most forces such as gravity and the normal force in fact obey the strong law of action and reaction.

§4.3 Force on a system of particles

Consider a system of N particles. For an ith particle, by the second law,

$$\left(\sum \mathbf{F}\right)_i + \sum_{j, j\neq i} \boldsymbol{f}_{ij} = \dot{\boldsymbol{p}}_i$$

where $\mathbf{F}$ is the external force and $\boldsymbol{f}_{ij}$ is the internal force on i due to j. Summing this for all N particles,

$$\sum \left(\sum \mathbf{F}\right)_i + \sum_i \sum_{j, j\neq i} \boldsymbol{f}_{ij} = \sum_i \dot{\boldsymbol{p}}_i$$

Note that the first term is simply the net external force, $\sum \mathbf{F}_{ext}$. The second term actually goes to zero since if we split the sum into action-reaction pairs, $\boldsymbol{f}_{ij} + \boldsymbol{f}_{ji}$, they cancel out by the third law.

So we simply have,

$$\sum \mathbf{F}_{ext} = \sum_{i=1}^{N} \dot{\boldsymbol{p}}_i$$

It's very difficult to compute the sum of momentums for large values of N. Thus we define something called the *centre of mass*.

Definition 4.2

The position vector, $\mathbf{R}$, of the Centre of Mass of a (system)body of N particles is,

$$\mathbf{R} = \frac{\sum_{i=1}^{N} m_i \boldsymbol{r}_i}{\sum_{i=1}^{N} m_i} \tag{4.1}$$

If the body has a mass M,

$$M = \sum_{i=1}^{N} m_i,$$

Then, for infintisemal $d\boldsymbol{r}$,

$$M\,d\boldsymbol{R} = \sum_{i=1}^{N} m_i\,d\boldsymbol{r_i}$$

Note that,

$$\sum_{i=1}^{N} \boldsymbol{p}_i = \sum_{i=1}^{N} m_i \frac{d\boldsymbol{r}_i}{dt}$$

Thus,

$$\boxed{\sum_{i=1}^{N} \boldsymbol{p}_i = M\dot{\mathbf{R}}} \tag{4.2}$$

Where let $\boldsymbol{R}$ is the position vector of the centre of mass(CM). Thus,

$$\sum_{i=1}^{N} \boldsymbol{p}_i = M\boldsymbol{v}_{CM}$$

And,

$$\sum_{i=1}^{N} \dot{\boldsymbol{p}}_i = M\boldsymbol{a}_{CM}$$

Which finally gives us,

$$\boxed{\sum \mathbf{F}_{ext} = M\boldsymbol{a}_{CM}.} \tag{4.3}$$

This means that a system of particles evolves translationally due to a net external force as if it were a point particle of mass M located at the center of mass of the system, with a position vector $\mathbf{R}$.

This means that we can simply analyze the translational motion of the center of mass which defines the translational motion of the entire rigid body, as the position vector of the particles are fixed relative to each other.

For continous mass distributions,

$$\mathbf{R} = \frac{\int \boldsymbol{r}\,dm}{\int dm} \tag{4.4}$$

§4.3.1 Calculating the Centre of Mass

A good reference here is [WR18].

Let us first calculate the centre of mass of two particles of masses m and M located at (x_1, y_1, z_1) and (x_2, y_2, z_2). By definition 4.2,

$$\mathbf{R} = \frac{1}{m+M}\left[\begin{pmatrix} mx_1 \\ my_1 \\ mz_1 \end{pmatrix} + \begin{pmatrix} Mx_2 \\ My_2 \\ Mz_2 \end{pmatrix}\right] = \frac{1}{m+M}\begin{pmatrix} mx_1 + Mx_2 \\ my_1 + My_2 \\ mz_1 + Mz_2 \end{pmatrix}$$

One special thing we may note is that instead of calculating the centre of mass for each particle, we reduce some say n particles or distributions into point particles by first calculating their centre of mass. For instance, instead of calculating the com of three them at once, we can calculate the com of the first two, and then that of the third.

This works because suppose the com of any n particles or distributions is located at $\mathbf{R}$. Then, if we calculate the com of this with another say $n+1$th particle,

$$\mathbf{R}' = \frac{M\mathbf{R} + m_{n+1}\boldsymbol{r}_{n+1}}{M + m_{n+1}},$$

where M is the total mass of the n particles. By the definition of com,

$$M\mathbf{R} = \sum_{i=1}^{N} m_i \boldsymbol{r}_i$$

Replacing this,

$$\mathbf{R}' = \frac{\sum_{i=1}^{N} m_i \boldsymbol{r}_i + m_{n+1}\boldsymbol{r}_{n+1}}{\sum_{i=1}^{N} m_i + m_{n+1}},$$

Which is simply the com of the $n+1$ particles, at once calculated!

$$\mathbf{R}' = \frac{\sum_{i=1}^{N+1} m_i \boldsymbol{r}_i}{\sum_{i=1}^{N+1}}$$

The same can be extended to mass distributions.

For example, to calculate the com of two spheres of uniform mass distribution, we may calculate their com, which is at their centre (by a symmetry argument), and then just use our previous result of com of two point masses.

This can also be used to find the com for "missing masses". Where we replace our incomplete system with a complete system having the total mass of the system if it were complete, and a negative mass. This might be a little difficult to understand, so let us look at an example:

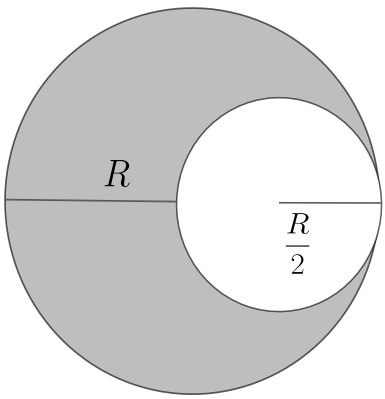


Figure 4.1: Missing mass

Example 4.3 A uniform circle of radius R has a circular hole of radius $R/2$. The surface density of the circle is σ. Find the centre of mass of the system.

Solution. To calculate the com, we first "fill up" or replace the incomplete the circle with a complete circle. This can then be reduced to a point mass of mass $M = \sigma\pi r^2$.

To counter effect this addition of mass by filling up the circle, we associate a negative mass with the smaller circle, $-m = -\frac{\sigma\pi r^2}{4}$.

Let the origin the com of the first circle, then the first circle, which we have reduced to be a particle at its com, has a position vector, $\mathbf{0}$. Thus, com of the system is,

$$\mathbf{R} = \frac{M \times 0 - mR/2}{M - m} = \frac{-mR/2}{M - m} = \frac{R}{6}$$

□

For continuous distributions, we may use integration to calculate the centre of mass,

Example 4.4 Determine the center of mass of a right-angled triangle of a uni- form surface mass density σ and with an angle of inclination in the xy plane. Let the length of its base be ℓ.

Solution. Note that $\sigma = dm/\,dA \iff dm = \sigma\,dA = \sigma\,dx\,dy$.

So, we can setup a double integral,

$$\mathbf{R} = \frac{1}{M}\int_0^{\ell}\int_0^{x\tan\theta} x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}}\sigma\,dy\,dx$$

Splitting up the integrals,

$$\begin{aligned}
R_x &= \frac{1}{M}\int_0^{\ell}\int_0^{x\tan\theta} x\sigma\,dy\,dx \\
R_x &= \frac{1}{M}\int_0^{\ell} \sigma\,xy\Big|_0^{x\tan\theta}\,dx \\
R_x &= \frac{1}{M}\int_0^{\ell} x^2\tan\theta\sigma\,dx \\
R_x &= \frac{1}{M}\,\frac{x^3\sigma\tan\theta}{3}\Big|_0^{\ell} \\
R_x &= \frac{2}{\ell^2\sigma\tan\theta}\,\frac{\ell^3\sigma\tan\theta}{3} \\
R_x &= \frac{2\ell}{3}
\end{aligned}$$

And by a similar procedure, $R_y = \ell\tan\theta/3$. Which gives you $\mathbf{R}$.

□

§4.4 Some Phenomenological Forces

§4.4.1 The Normal Force

The contact force on a body can be resolved to two mutually perpendicular forces, the normal force, perpendicular to the object, and friction, tangential to the object.

These contact forces arise due to atomic interactions. If we say keep a book on a table, the book pushes against the table. On a microscopic scale, the molecules push against each other. After a certain point, the molecules start repulsing each other and the table applies a normal force, perpendicular to the body, to stop it.

Since no object is perfectly rigid, there always occurs a deformation on its surface. However, this is usually negligible, and we may safely assume that the object is perfectly rigid.

§4.4.2 Friction

Friction much like the normal force, arises from molecular interactions. It is, however, way too complex to analyse. Thus, our study of friction is pure phenomenological. Here, by friction we refer to dry friction, and not as is, drag.

Friction opposes the relative motion of objects — and this is in fact the most we can say safely, and generally. It is independent of the contact area.

This might feel a little weird, but consider the following argument — say we have an object who has contact area with another object. This contact area is actually a very small fraction of its overall surface area. It is, also, proportional to the pressure.

If we double the surface area of the object, let us assume that the contact area doubles as well. If we keep the normal force constant, the pressure halves. Friction, however, the product of the now half pressure, and the double contact area, remains the same.

This is a very ideal approximation of friction, but we'll leave it here for now.

We differentiate between static friction and kinetic friction. When we apply a force on an object, and yet it does not undergo motion, the static friction assumes a value to balance out the force. This friction is actually related to the normal by $|f_s| \leqslant \mu_s N$. Where μ_s is the coefficient of static friction.

However, it can not keep the object at rest forever, after a certain amount of force is applied, the object undergoes motion and experiences kinetic friction. It is approximately constant and related to the normal by $|f_k| = \mu_k N$, where μ_k is the coefficient of kinetic friction.

Generally, $\mu_k < \mu_s$ and they're both less than 1. Though for some exceptionally rough surfaces, they may have a value greater than 1.

§4.4.3 Spring Force

The relation between the spring force and the displacement of the spring from its equilibrium position x_0 was empirically found by Robert Hooke and is popularly known as Hooke's Law, which states that,

$$F_s = -k(x - x_0) \tag{4.5}$$

where x is the displacement from its equilibrium position, x_0. k is the spring constant and depends on the spring used.

Hooke's law actually fails for rather large displacements but is remarkably good for small displacements. It works not only here, but in a lot more places where the restoring forces are concerned.

The springs can be connected in two main ways.

§4.4.4 Tension

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